Statistical Learning Exercices 1

Exercice 1. *Assume that we have a set of data that describes the following three variables:* $Hair = \{B, D\}$, *where* $B = blond, D = dark$. $Height = \{T, S\}$, *where* $T = tall, S = short$. $Country = \{G, P\}$, *where* $G = Greenland, P = Poland$. *You are given the following workout data set (Hair, Height, Country):*

$$\begin{pmatrix} (B,T,G) & (B,T,G) & (B,T,P) \\ (D,T,G) & (D,T,G) & (B,T,P) \\ (D,T,G) & (D,T,G) & (B,T,P) \\ (D,T,G) & (D,T,G) & (D,T,P) \\ (B,T,G) & (B,T,G) & (D,T,P) \\ (B,S,G) & (B,S,G) & (D,S,P) \\ (B,S,G) & (B,S,G) & (B,S,P) \\ (D,S,G) & (D,S,G) & (D,S,P) \end{pmatrix}$$

We want to answer the following question: if you observe a new tall individual having blond hair, what is her most likely country of origin?

1. *Give the maximum posterior solution (MAP) to the question, using the naive frame from Bayes. Detail your work.*
2. *Give the maximum a posteriori solution (MAP) to the question, without using the naive frame from Bayes. Detail your work.*
3. *Give the maximum likelihood solution (MLE) to the question, using the naive frame from Bayes. Detail your work.*
4. *Give the maximum likelihood solution (MLE) to the question, without using the naive frame from Bayes. Detail your work.*

Exercice 2. *We consider a tiny data set with* 10 *observations* $(x_i, y_i)_{1 \leq i \leq 10}$, *with* $x_i \in \mathcal{X}$ *and* $y_i \in \{-1, 1\}$. *Using different machine learning algorithms, two models are built from this data set:* g_1 *and* g_2. *Recall that a model is a function from* $\mathcal{X}$ *to* $\{-1, 1\}$. *The outputs of the models on the learning data set are given by the following table:*

x_i	$g_1(x_i)$	$g_2(x_i)$	y_i
x_1	1	1	1
x_2	1	-1	1
x_3	-1	1	1
x_4	1	-1	1
x_5	1	-1	1
x_6	1	-1	-1
x_7	-1	-1	-1
x_8	-1	-1	-1
x_9	-1	1	-1
x_{10}	1	-1	-1

1. *Compute the confusion matrices of both models on the learning set.*

2. *We use the loss function ℓ_1 given by:*

$\ell_1(p,t)$	*t=-1*	*t=1*
p=-1	*0*	*1*
p=1	*3*	*0*

where p is the predicted value and t the true value. Compute the empirical risk of both models on the learning set for ℓ_1.

3. *Determine the best model based on the available information using the loss function $L_b(p,t) = \mathbf{1}_{p \neq t}$.*